

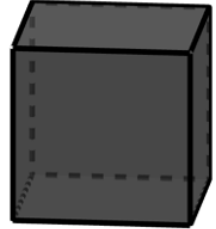
Grade 6 Accelerated
Unit 8
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

A $\frac{1}{3}$ inch unit cube is shown.

1. Determine the volume of the unit cube. Include the units for the cube's volume.

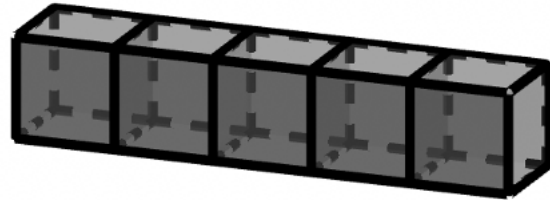
$$\frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{27} \text{ in}^3$$



The diagram shows five unit cubes whose dimensions are $\frac{1}{5}$ inches.

2. The volume of one cube is
 $\frac{1}{125}$ cubic inches (in^3).

$$\frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{5} = \frac{1}{125} \text{ in}^3$$



3. The volume of all five cubes is $\frac{1}{25}$ in^3 .

$$5 \cdot \frac{1}{125} = \frac{1}{25}$$

A rectangular prism is shown.

4. How many unit cubes whose dimensions are $\frac{1}{4}$ inch fill up the rectangular prism?

16

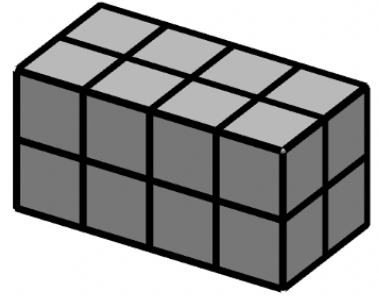
5. Each unit cube whose dimensions are $\frac{1}{4}$ inch has a volume of $\frac{1}{64}$ cubic inches. Find the total volume, in cubic inches, represented in the diagram by multiplying the number of cubes that fill the rectangular prism by the volume of one of the cubes.

$$16 \cdot \frac{1}{64} = \frac{1}{4} \text{ in}^3$$

6. Find the total volume represented in the diagram, in cubic inches, using the formula, $V = lwh$.

$$l = 1 \quad w = \frac{1}{2} \quad h = \frac{1}{2}$$

$$V = 1 \cdot \frac{1}{2} \cdot \frac{1}{2} \quad V = \frac{1}{4} \text{ in}^3$$



A rectangular prism with one cube inside is shown.

7. How many cubes will it take to fill the rectangular prism?
3 cubes across, 3 cubes back, 4 cubes high
36 cubes

8. The rectangular prism has a length of $\frac{12}{5}$ centimeters (cm), a width of $\frac{12}{5}$ cm, and a height of $\frac{16}{5}$ cm as shown. Determine the dimensions of the fractional unit cube.

$$\frac{16}{5} \div 4 = \frac{16}{5} \cdot \frac{1}{4} = \frac{4}{5} \text{ cm}$$

$$\frac{12}{5} \div 3 = \frac{12}{5} \cdot \frac{1}{3} = \frac{4}{5} \text{ cm}$$

Each cube has dimensions of $\frac{4}{5}$ cm.

9. Determine the volume of the unit cube.

$$\frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5} = \frac{64}{125} \text{ cm}^3$$

10. Determine the volume of the rectangular prism.

$$36 \cdot \frac{64}{125} = \frac{2,304}{125} = 18 \frac{54}{125} \text{ cm}^3$$

